

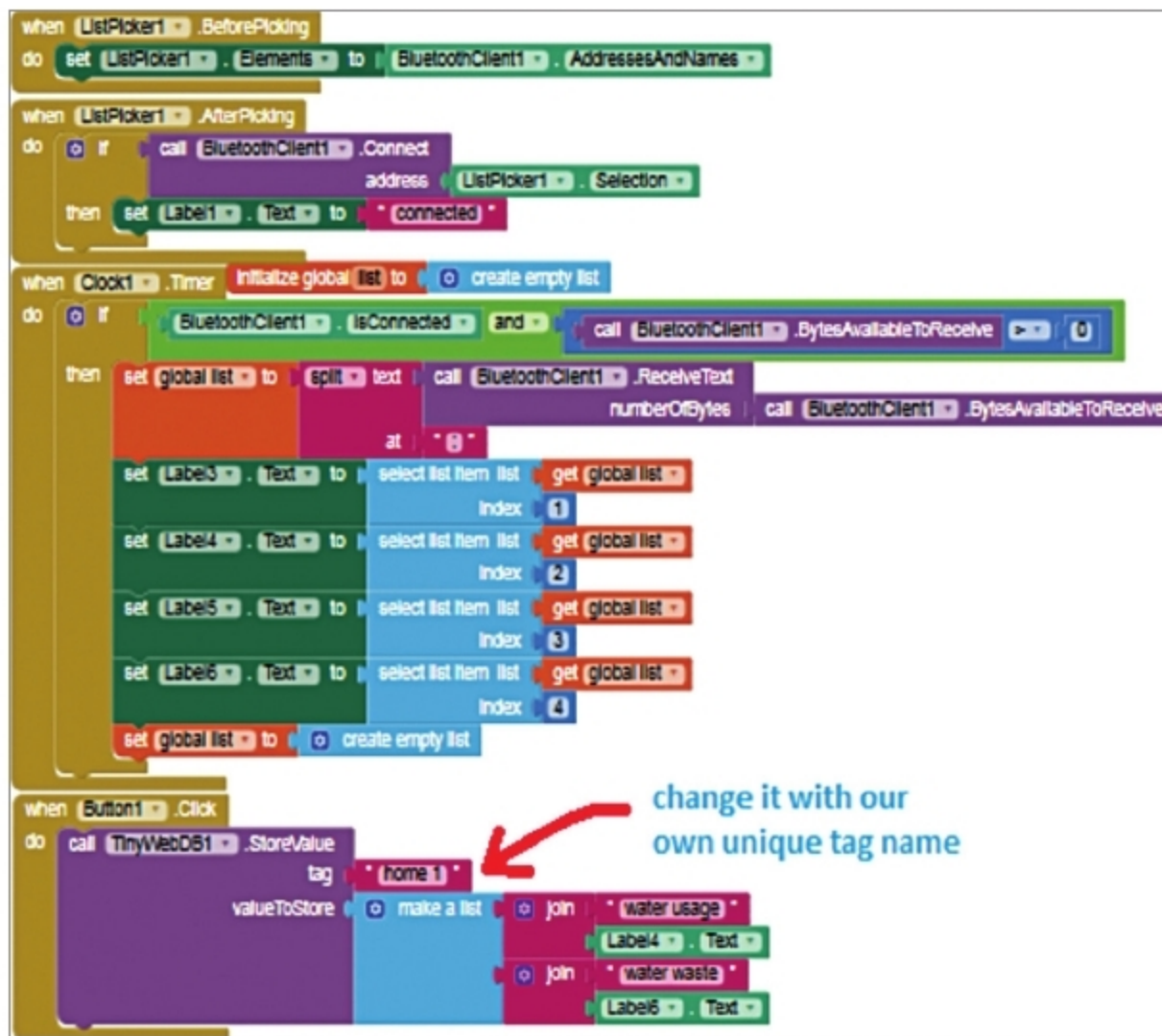


Fig. 9: MIT App code blocks coding

Connections Between Arduino Uno and Pro Mini Boards Table 1

Arduino Uno	Arduino Pro Mini
Rx	Rx1
Tx	Tx0
Gnd	Gnd
5V	Vcc
Reset	Reset

Arduino Pro Mini Pins Connections With Components Table 2

Arduino Pro Mini pins	Components pins
5V	Vcc of Bluetooth
GND	GND of Bluetooth
RX1	TX of Bluetooth
TX0	RX of Bluetooth
Vcc	OLED Vcc
GND	OLED Vcc
A4(SDA)	OLED SDA
A5(SCL)	OLED SCL
Pin 3	Output pin of water flow sensor

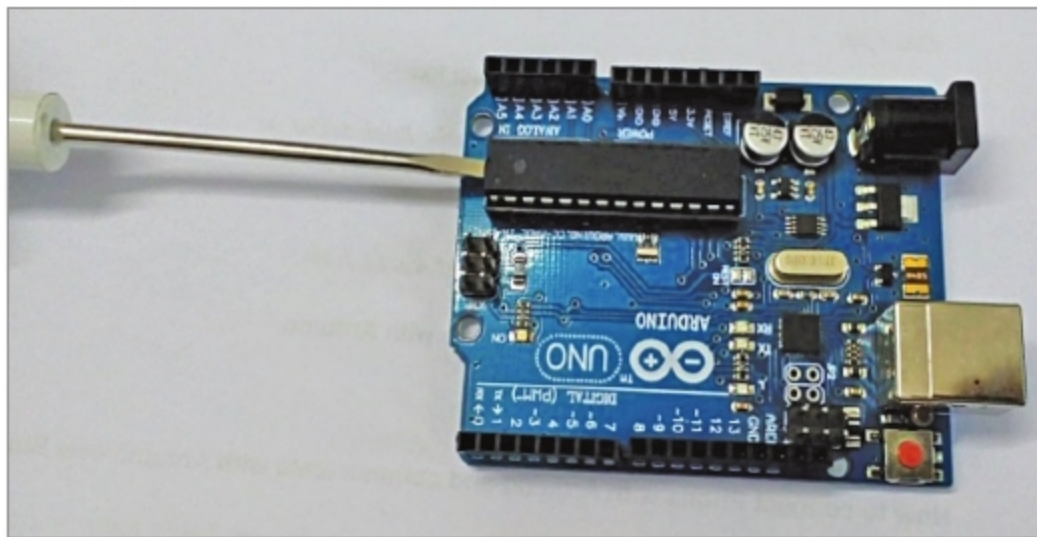


Fig. 10: Removing ATmega328 IC from the board

tions with the components are given in Table 2. The actual wiring diagram of the Arduino Pro Mini pins with the components is shown in Fig. 12.

Now, cross-check all the connections. If all the connections are correct, then power the Arduino with a 5V DC power supply.

Connect the water sensor pipe to the water tap or water inlet of your water tank. You can see the total volume of water and water flow rate on the OLED display. Next, install the smart_meter.apk file on your Android device. Then, open smart_meter app and press on the Bluetooth icon.

You will get a list of Bluetooth

devices available near your Android phone. Select HC-05 from the list to pair it. The default code for pairing is 1234 (or could be 0000 or 1111). After connecting/pairing it, you will get the water flow rate and water volume consumed in the app on your phone as shown in Fig. 13. If there is a display problem or the data is incorrect, disconnect Bluetooth and connect it again.

You can tap on the DB icon to upload your water usage data to the Web server so that it can be seen from anywhere through the Internet.

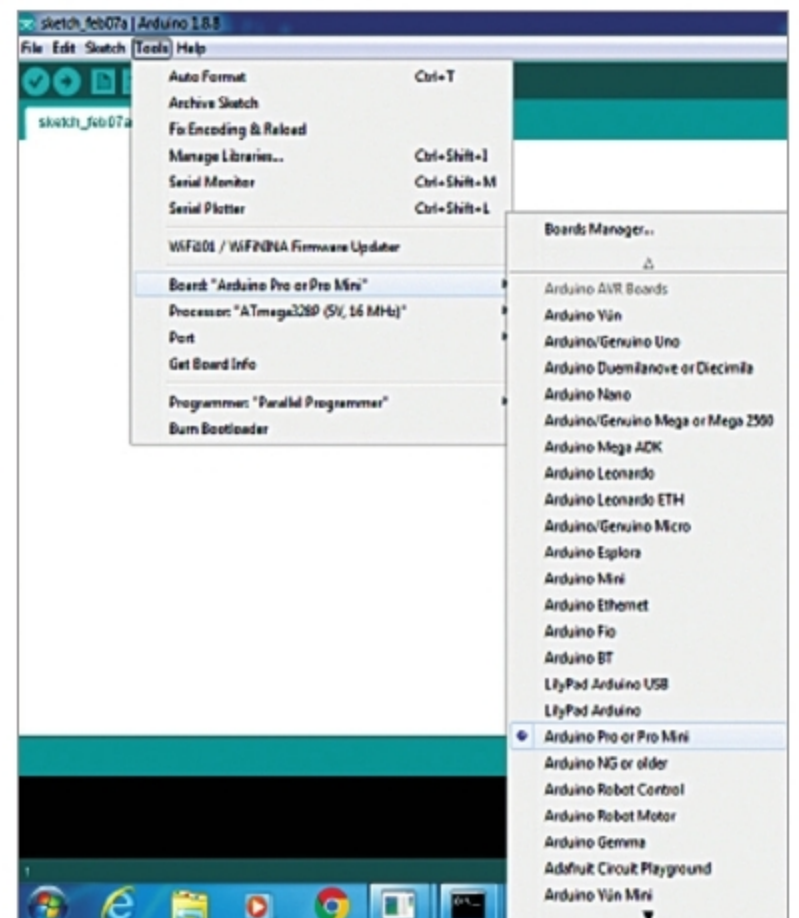


Fig. 11: Selecting the Board in Arduino IDE

Note. Internet Wi-Fi connection needs to be turned on in order to upload the data to the server.

Now, to fetch the data, click on the link <http://appinvtinywebdb.appspot.com/getvalue>. Then type your house tag name as home 1, which is already mentioned in the app. Click on Get